

TURNING POINT

ASSAULT RIFLES

Just as the breech-loading repeating rifle had brought about a change in warfare following its introduction in the late 19th century, the development of reliable self-loading military arms during the 1930s altered tactics again—now a single infantryman could deliver fire equivalent to a squad of 10 or 12. In 1944, the assault rifle magnified this effect almost 50-fold as it mimicked a machine-gun. Easy to use, an assault rifle allowed anyone to become an effective adversary, transforming warfare from a clash between trained armies on a battlefield to a contest between masses, often street-to-street or even house-to-house.



▲ ASSAULT RIFLE

An assault rifle is a short-barreled rifle, intended for use by infantry, and capable of selective fire—switching between semiautomatic and automatic modes. It chambers medium- and small-caliber cartridges with short cases. It has a high-capacity magazine that can carry 20 or more rounds. Shown here is a 1954 AK47, which fires 7.62 × 39mm cartridges.

Conflicts at the turn of the 20th century saw the development of groundbreaking weaponry. Firearms were modernized with the invention of the Maxim gun—the first machine-gun (see pp.184–85)—which spurred the refinement of automatic weapons technology at a furious pace. Heavy machine-guns were followed by medium- and light machine-guns, as armies felt need to provide groups of soldiers with portable, automatic firepower. It was not until the invention and use of the assault rifle during World War II (1939–45) that this deadly objective was fully achieved.

EARLY EXPERIMENTS

The precursor to the modern assault rifle—Burton's automatic rifle of 1917—had twin, 20-round magazines for use by a single rifleman. It chambered short-cased, high-velocity cartridges and was a selective-fire weapon—it could be used as a single-shot,



▲ 5.56 × 45MM AND 7.62 × 51MM CARTRIDGES

To prevent heavy recoil, assault rifles fire short-cased, small-caliber or “intermediate” cartridges (left) instead of long-cased, large-caliber rifle cartridges (right).

self-loading arm or fired in bursts like a machine-gun. Except for its barrel length, it matched all the modern criteria for a weapon to be deemed an assault rifle. However, the design was ahead of its time and was never adopted for production. The first mass-produced assault rifle was the German Sturmgewehr 44, or StG44 (see p.176). It was used extensively in World War II on both the Eastern and Western fronts and provided the German troops with an effective countermeasure to the Soviet submachine-gun, the PPSH-41 (see p.208). Between 1945 and 1946, Soviet arms dealer Mikhail Kalashnikov designed a modern assault rifle, and in 1947, he unveiled the AK47 (see pp.248–49).

MODERN ASSAULT RIFLES

The AK47 embodied all the features typical of assault rifles; it had a short barrel, a high-capacity magazine, and full- and semiautomatic fire controls. In the West, development of the assault

“I created a weapon to defend the borders of my motherland. It's not my fault that it's being used where it shouldn't be...”

MIKHAIL KALASHNIKOV,
SOVIET AK47 DESIGNER

KEY FIGURE

Frank F. Burton
(1871–1939)

Frank F. Burton was the son of the famed civil engineer Col. James Henry Burton. He joined the Winchester Repeating Arms Company as a designer in the 1890s. He designed his assault rifle in response to a need for a light automatic arm for observers in aircrafts prior to the introduction of synchronized machine-guns.



» BEFORE

Prior to the development of the assault rifle, concentrated fire in volume could only be delivered by machine-guns. Their long medium-caliber rounds were capable of accuracy at up to 3,000ft (900m).

● **SOME LIGHT MACHINE-GUNS**, such as the 1918 Browning Automatic Rifle (BAR), were intended to replace heavy machine-guns for small groups of soldiers. However, they were heavy and unwieldy.

● **SUBMACHINE-GUNS** were intended to be an ideal replacement for the machine-gun. In practice though, their reliance on pistol cartridges meant that they were effective only at close range and were not able to fulfill the functions of a multipurpose combat weapon.



BURTON'S
AUTOMATIC RIFLE

● **BURTON'S AUTOMATIC RIFLE**, designed in 1917, was the ancestor of assault rifles. It used a .345-caliber cartridge and was capable of selective fire.



BROWNING
AUTOMATIC RIFLE